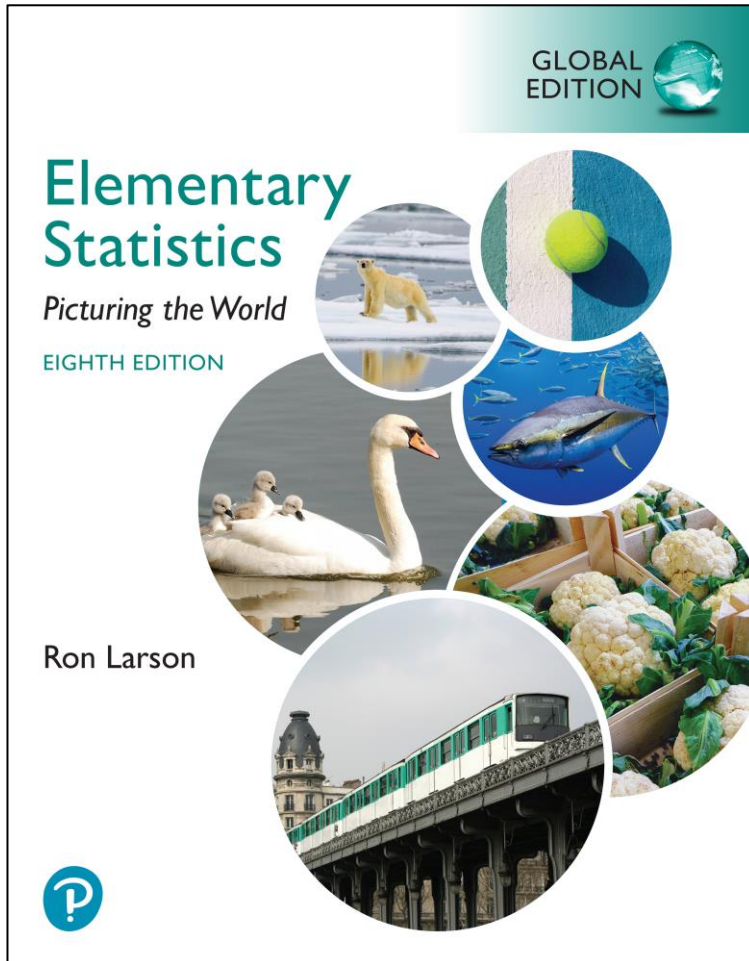


Elementary Statistics

Eighth Edition



Chapter 4

Discrete Probability Distributions

Chapter Outline

4.1 Probability Distributions

4.2 Binomial Distributions

4.3 More Discrete Probability Distributions

Section 4.3 More Discrete Probability Distributions

Section 4.3 Objectives

1. How to find probabilities using the geometric distribution
2. How to find probabilities using the Poisson distribution

Geometric Distribution

Geometric distribution

- A discrete probability distribution.
- Satisfies the following conditions
 1. A trial is repeated until a success occurs.
 2. The repeated trials are independent of each other.
 3. The probability of success p is constant for each trial.
 4. The random variable x represents the number of the trial in which the first success occurs.
- The probability that the first success will occur on trial x is

$$P(x) = p(q)^{x-1}, \text{ where } q = 1 - p.$$

Example: Using the Geometric Distribution

A recent study found that the failure rate of businesses after five years is 50%. Four businesses that started five years ago are selected at random. Find the probability that the fourth business selected is the first one to have failed. (Source: U.S. Bureau of Labor Statistics)

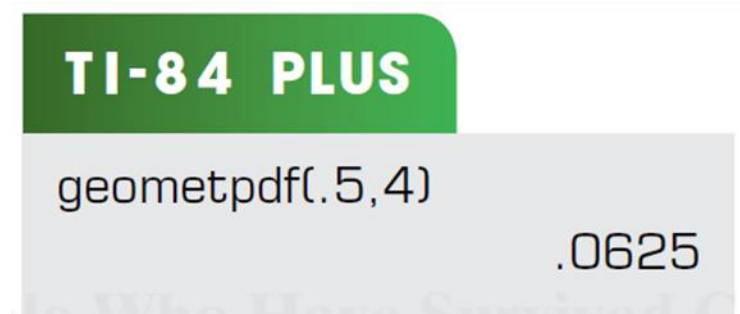
Solution: Using the Geometric Distribution

Solution

Using $p = 0.50$, $q = 0.50$, and $x = 4$, you have

$$\begin{aligned}P(4) &= 0.50(0.50)^{4-1} \\&= 0.50(0.50)^3 \\&\approx 0.063.\end{aligned}$$

So, the probability that the fourth business selected is the first one to have failed is about 0.063.



Poisson Distribution (1 of 2)

Poisson distribution

- A discrete probability distribution
- Satisfies the following conditions
 1. The experiment consists of counting the number of times an event, x , occurs in a given interval. The interval can be an interval of time, area, or volume.
 2. The probability of the event occurring is the same for each interval.
 3. The number of occurrences in one interval is independent of the number of occurrences in other intervals.

Poisson Distribution (2 of 2)

Poisson distribution

Conditions continued:

- The probability of exactly x occurrences in an interval is

$$P(x) = \frac{\mu^x e^{-\mu}}{x!} \quad \text{where } e \approx 2.71818 \text{ and } \mu$$

is the mean number of occurrences

Example: Using the Poisson Distribution

The mean number of accidents per month at a certain intersection is three. What is the probability that in any given month four accidents will occur at this intersection?

Solution:

- Poisson with $x = 4$, $\mu = 3$

$$P(4) = \frac{3^4 (2.71828)^{-3}}{4!} \approx 0.168$$



TI-84 PLUS

poissonpdf(3,4)

.1680313557

Example: Finding a Poisson Probability Using a Table

A population count shows that the average number of rabbits per acre living in a field is 3.6. Use a table to find the probability that seven rabbits are found on any given acre of the field.

Solution: Finding a Poisson Probability Using a Table (1 of 2)

Solution:

A portion of Table 3 in Appendix B is shown here. Using the distribution for $\mu = 3.6$ and $x = 7$, you can find the Poisson probability as shown by the highlighted areas in the table.

	μ						
x	3.1	3.2	3.3	3.4	3.5	3.6	3.7
0	.0450	.0408	.0369	.0334	.0302	.0273	.0247
1	.1397	.1304	.1217	.1135	.1057	.0984	.0915
2	.2165	.2087	.2008	.1929	.1850	.1771	.1692
3	.2237	.2226	.2209	.2186	.2158	.2125	.2087
4	.1734	.1781	.1823	.1858	.1888	.1912	.1931
5	.1075	.1140	.1203	.1264	.1322	.1377	.1429
6	.0555	.0608	.0662	.0716	.0771	.0826	.0881
7	.0246	.0278	.0312	.0348	.0385	.0425	.0466
8	.0095	.0111	.0129	.0148	.0169	.0191	.0215
9	.0033	.0040	.0047	.0056	.0066	.0076	.0089
10	.0010	.0013	.0016	.0019	.0023	.0028	.0033

Solution: Finding a Poisson Probability Using a Table (2 of 2)

According to the table, the probability is 0.0425. You can check this result using technology. As shown below using Excel, the probability is 0.042484.

After rounding to four decimal places, the probability is 0.0425, which is the same value found using the table.

EXCEL		
	A	B
1	POISSON(7,3.6,FALSE)	
2		0.042484